

NEWTON'S LAW OF GRAVITATION

$$F = \frac{Gm_1m_2}{r^2}$$

G - Universal gravitational constant
Value of G

$$6.67 \times 10^{-11} \text{ Nm}^2\text{Kg}^{-2} \text{ (SI or MKS)}$$

$$6.67 \times 10^{-8} \text{ dyne cm}^2\text{g}^{-2} \text{ (CGS)}$$

Dimensional formula G
[G] = [M⁻¹L³T⁻²]

ROTATION OF 2 MASSES UNDER MUTUAL GRAVITATIONAL FORCE OF ATTRACTION

$$V_1 = \sqrt{\frac{Gm_2^2}{(m_1+m_2)r}}$$

$$V_2 = \sqrt{\frac{Gm_1^2}{(m_1+m_2)r}}$$

$$\text{Time period } T^2 = \frac{4\pi^2}{G(m_1+m_2)} r^3$$

or $T^2 \propto r^3$

VARIATION IN THE VALUE OF ACCELERATION DUE TO GRAVITY

• Variation due to height 'h'

$$g' = \frac{GM}{(R+h)^2}$$

$$g' = \frac{gR^2}{(R+h)^2}$$

General Equation

$$\text{Approximate equation } h \ll R \text{ (} h < 100 \text{ km)}$$

use, $g' = g \left(1 - \frac{2h}{R}\right)$

Note the point

If $h \ll R$, then decrease in the value of g with height

$$\text{Absolute decrease} = \Delta g = g - g' = \frac{2hg}{R}$$

$$\text{Fractional decrease} = \frac{\Delta g}{g} = \frac{g-g'}{g} = \frac{2h}{R}$$

$$\text{Percentage decrease} = \frac{\Delta g}{g} \times 100 = \frac{g-g'}{g} \times 100 = \frac{2h}{R} \times 100$$

• Variation due to depth 'd'

$$g' = g \left[1 - \frac{d}{R}\right]$$

$$= \frac{gr}{R}$$

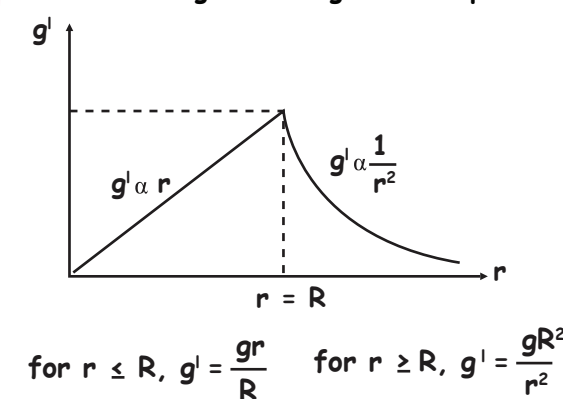
$$\text{Absolute decrease} = \frac{\Delta g}{g} = g - g' = \frac{dg}{R}$$

$$\text{Fractional decrease} = \frac{\Delta g}{g} = \frac{g-g'}{g} = \frac{d}{R}$$

$$\text{Percentage decrease} = \frac{\Delta g}{g} \times 100 = \frac{d}{R} \times 100$$

Very important graph

The graphical representation of change in the value of g' with height and depth



• Variation of g due to rotation of earth

Latitude - Angle which the line joining the point to the centre of earth makes with the equatorial plane

$$g' = g - \omega^2 R \cos^2 \lambda$$

Note $\Rightarrow$ value of $\omega^2 R = 0.034$

For poles $\lambda = 90^\circ$ $g' = g$

There is no effect of rotational motion of the earth on the value of g at poles.

For equator $\lambda = 0^\circ$ $g' = g - \omega^2 R$

The effect of rotational motion of the earth on the value of g at the equator is maximum.

When a body of mass m is moved from equator to the poles, weight increases by an amount $m(g_p - g_e) = m\omega^2 R$

WORK DONE IN MOVING OBJECT FROM ONE ORBIT TO ANOTHER

$$V_{01} = \sqrt{\frac{GM}{r_1}}$$

$$V_{02} = \sqrt{\frac{GM}{r_2}}$$

CONCEPT - WORK DONE BY EXTERNAL AGENT = CHANGE IN MECHANICAL ENERGY

$$W = E_2 - E_1 = \frac{GMm}{2} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

IMPORTANT POINTS ABOUT GRAVITATIONAL FORCE

1. Gravitational force is
 - * Always attractive in nature
 - * Independent of the nature of medium between masses
 - * Independent of presence or absence of other bodies
2. Is a central force, acts along the line joining centre of gravity of two bodies.
3. Conservative force

THREE EQUAL MASSES REVOLVING UNDER MUTUAL GRAVITATIONAL FORCE

$$V_1 = \sqrt{\frac{GM}{l}} = \sqrt{\frac{GM}{R\sqrt{3}}}$$

$$\text{Time period } T^2 \propto l^3$$

FOUR EQUAL MASSES UNDER MUTUAL GRAVITATIONAL FORCE

$$V = \sqrt{\frac{GM}{2\sqrt{2}l}} (2\sqrt{2}+1)$$

$$V = \frac{1}{2} \sqrt{\frac{GM}{R}} (2\sqrt{2}+1)$$

$$T^2 \propto l^3 \text{ Here } R = \frac{l}{\sqrt{2}}$$

GRAVITY

Acceleration due to gravity on the surface of earth, $g = \frac{GM_e}{R_e^2}$

M - mass of earth
R - Radius of earth
[Put $GM_e = gR_e^2$ to solve problems easily]

g IN TERMS OF DENSITY OF EARTH

$$g = \frac{4}{3} \pi G \rho R_e \quad g \propto \rho R_e$$

"If density is mentioned use the above equation"

VECTOR FORM

$\vec{F}_{21}$ = force on 2 due to 1
 $\vec{r}_{12}$ = vector from m_1 to m_2

$$\vec{F}_{21} = \frac{Gm_1m_2}{r_{12}^2} (-\hat{r}_{12})$$

$$\vec{F}_{21} = -\frac{Gm_1m_2}{r_{12}^3} \vec{r}_{12}$$

Similarly

$$\vec{F}_{12} = \text{force on 1 due to 2}$$

$$\vec{F}_{12} = \frac{Gm_1m_2}{r_{12}^2} (\hat{r}_{12}) \text{ or } \vec{F}_{12} = \frac{Gm_1m_2}{r_{12}^3} \vec{r}_{12}$$

Clearly, Gravitational force follows:

Newtons third law $\vec{F}_{21} = -\vec{F}_{12}$

Gravitational force is a two body interaction. Force between two particles does not depend on the presence or absence of other particles. The principle of superposition is valid here. "Force on a particle due to a no. of particles is the resultant of forces due to individual particles."

ORBITAL VELOCITY OF A SATELLITE

Orbit at a height 'h' from the surface

$$V_o = \sqrt{\frac{GM}{(R+h)}} = \sqrt{\frac{gR^2}{(R+h)}}$$

If orbit is closer to earth's surface (neglect 'h') $V_o = \sqrt{\frac{GM}{R}} = \sqrt{gR}$

(called first cosmic velocity)

Note - for easy calculations

$$\sqrt{gR} = 8 \text{ km/s or } \sqrt{\frac{GM}{R}} = 8 \text{ km/s} = 8 \times 10^3 \text{ m/s}$$

$$\text{or } \frac{GM}{R} = 64 \times 10^6$$

K.E. P.E AND T.E FOR AN ORBITING SATELLITE

$$KE = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}$$

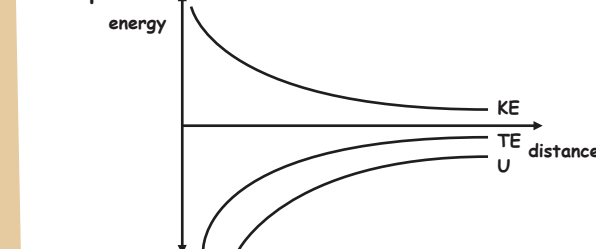
$$TE = -\frac{GMm}{2r}$$

Relation between K.E, U & T.E

$$U = 2 \times T.E$$

$$K.E = -T.E$$

Graph



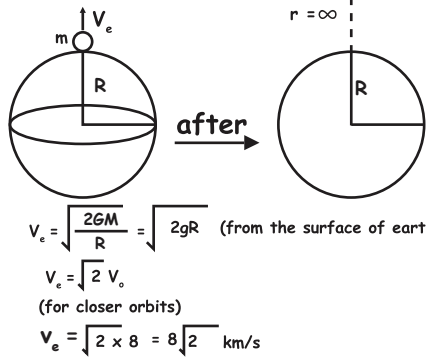
WORK DONE IN MOVING OBJECT FROM SURFACE TO CIRCULAR ORBIT

$$W = E_f - E_i$$

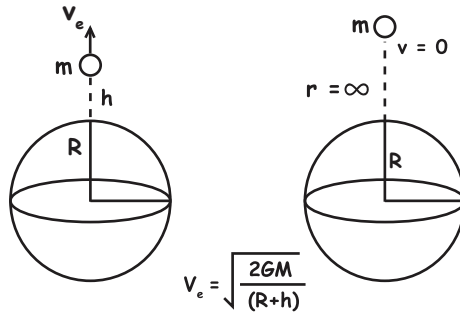
$$W = E_{\text{total}} - U_i = \frac{-GMm}{2(R+h)} + \frac{GMm}{R}$$

ESCAPE VELOCITY

"Minimum velocity given to an object such that it escapes out of Earth's gravitational field" $v=0$ $\circ$ m



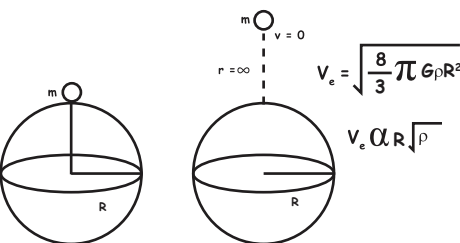
ESCAPE VELOCITY FROM A HEIGHT 'H'



ESCAPE ENERGY FOR ORBITING BODY

$$\Delta E = \text{Escape energy} = \frac{GMm}{2(R+h)}$$

ESCAPE VELOCITY FROM SURFACE OF EARTH AND RELATION WITH DENSITY



TRICK TO SOLVE PROBLEMS

Given speed greater than escape speed(Hint) find the final speed after escaping (question) short trick

if $V_{\text{given}} = nV_e$ (when $n > 1$)

final speed, $V = V_e \sqrt{n^2 - 1}$

Given speed less than escape speed(Hint) find the maximum height it reached (question) short trick

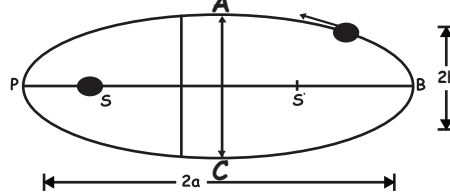
if $V_{\text{given}} = nV_e$ ($n < 1$)

maximum height,

$$h = \frac{n^2 R}{1 - n^2}$$

KEPLER'S LAWS OF PLANETARY MOTION

I) THE LAW OF ORBITS

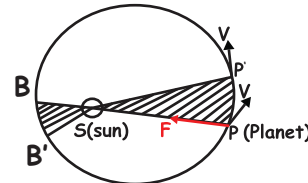


Every planet moves around the sun in an elliptical orbit with sun at one of the foci.

P → Perihelion (perigee) (nearest point)

B → apogee or aphelion (farthest point)

II) THE LAW OF AREAS



"The line joining the sun to the planet sweeps out equal areas in equal interval of time"

"i.e. areal velocity is constant"

"According to this law, planet will move slowly when it is farthest from sun & rapidly when is nearest to sun."

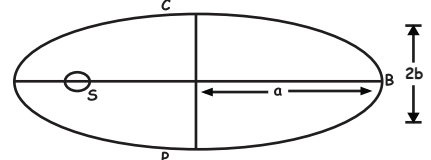
"Law of areas is due to law of conservation of angular momentum"

$$\text{Areal velocity} = \frac{L}{2m}$$

$$\frac{\Delta A}{\Delta t} = \frac{L}{2m} \quad L \rightarrow \text{Angular constant momentum}$$

⇒ Areal velocity is constant

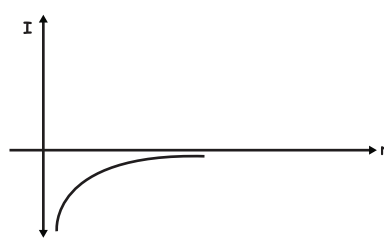
III) THE LAW OF PERIODS



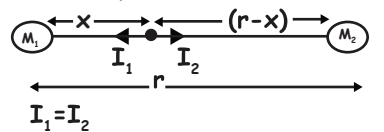
$$T^2 \propto a^3 \quad a \rightarrow \text{semi-major axis of elliptical orbit}$$

$$T^2 = \frac{4\pi^2}{GM_s} a^3 \quad M_s = \text{Mass of sun}$$

Graph



Neutral point

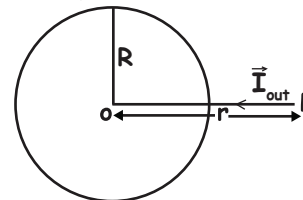


$$\frac{GM_1}{x^2} = \frac{GM_2}{(r-x)^2} \quad \text{or} \quad x = \frac{\sqrt{M_1} r}{\sqrt{M_1} + \sqrt{M_2}}$$

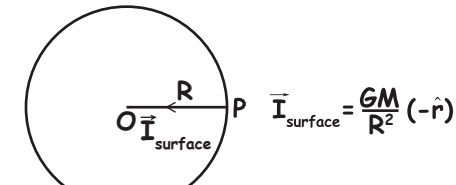
GRAVITATIONAL FIELD INTENSITY DUE TO A SPHERICAL SHELL

CASE-1 $r > R$

$$\vec{I}_{\text{out}} = \frac{GM}{r^2} (-\hat{r}) \Rightarrow I \propto \frac{1}{r^2}$$

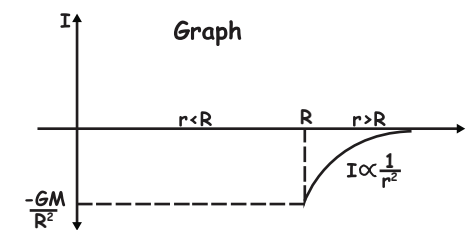


CASE-2 $r = R$



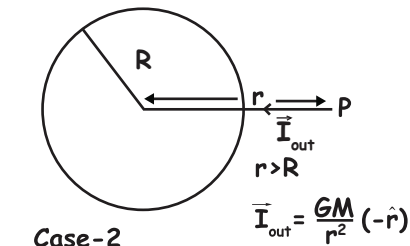
CASE-3 $r < R$

The point is inside then $I = 0$

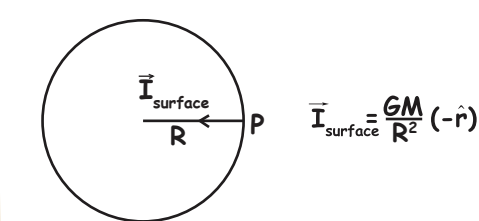


SOLID SPHERE

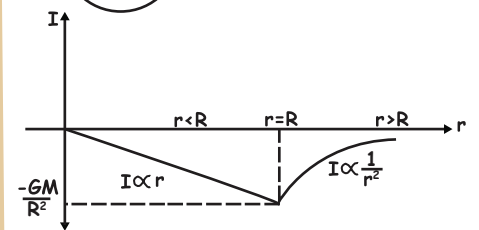
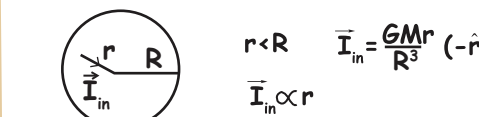
Case-1



Case-2



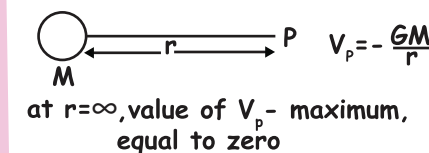
Case-3



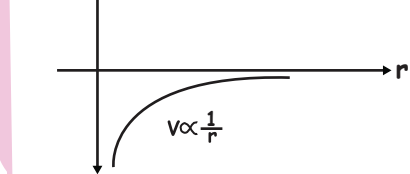
GRAVITATIONAL POTENTIAL

$$V = \frac{W_{\text{net}}}{m} \quad W_{\text{net}} = \text{Work done}$$

GRAVITATIONAL POTENTIAL FOR POINT MASS



at $r = \infty$, value of V_p - maximum, equal to zero



RELATION BETWEEN FIELD AND POTENTIAL

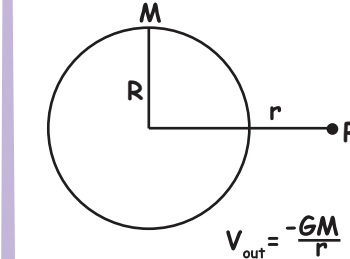
$$I = -\frac{dV}{dr} \quad \Delta V = \int \vec{I} \cdot d\vec{r}$$

$$\vec{I} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

GRAVITATIONAL POTENTIAL DUE TO OTHER BODIES

SOLID SPHERE

Case-1



Case-2

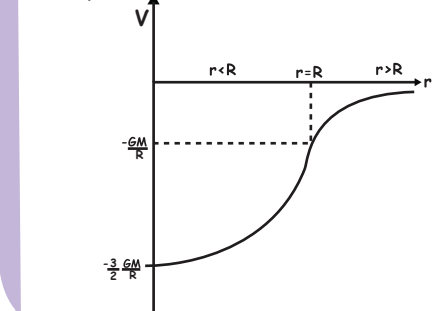
$$r = R \quad V_{\text{surface}} = -\frac{GM}{R}$$

Case-3

$$V_{\text{in}} = -\frac{GM}{2R^3} (3R^2 - r^2)$$

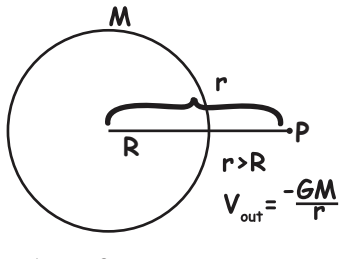
$$\text{At centre, } V_{\text{centre}} = -\frac{3}{2} \frac{GM}{R} = -\frac{3}{2} V_{\text{surface}}$$

Graph



HOLLOW SPHERE

Case-1



Case-2

$$r = R \quad V_{\text{surface}} = -\frac{GM}{R}$$

Case-3

$$V_{\text{in}} = V_{\text{surface}} = \text{same everywhere}$$

Graph

